DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
MONDAY TEST

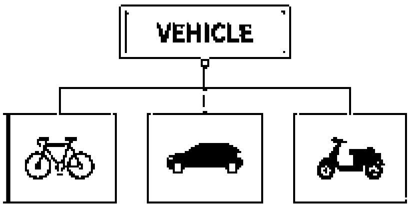
CLASS: X | **TOTAL MARKS: 40**
SUBJECT: COMPUTER APPLICATIONS | **DATE: 16.05.25**

Attempt all questions. The intended marks for questions are given in brackets [ ].
This paper consists of three printed pages.

Question 1 **[10x1=10]**

Choose the correct answers to the questions from the given options

(i)



Identify the feature of Java depicted by the above picture:
(a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Class And Object

(ii) **Which function is not overloaded in the Math class?**
(a) Math.max() (b) Math.pow() (c) Math.min() (d) Math.abs()

(iii) **Choose the correct output:**
System.out.println(Math.pow((Math.max(Math.ceil(-8.5),Math.floor(-8.8))),2));
(a) 81.0 (b) 81 (c) 72.0 (d) 64.0

(iv) **What is the return type of a method that does not return any value?**
(a) null (b)void (c) empty (d) zero

(v) **Assertion(A): Method signatures in Java include the method name and parameter list.**
Reason(R): Return type is not a part of the method signature in Java.
(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

(vi) **What will happen if a method that is supposed to return a value doesn't have the return statement?**
(a) Compilation error (b) Runtime error (c) No output (d) The method runs as void

(vii) **Choose the correct output:**

```
int i=1;
for ( ; i<= 3; i++);
System.out.print(i + " ");
```

(a) 123 (b)1 (c)1234 (d) 4

(viii) **Choose the correct output if n=4:**

```
String s = "3.14";
float f = Float.parseFloat(s);
System.out.println(f + 1);
```

(a) 4.14 (b) Error (c)3.141 (d) 3.14+1

(ix) **How many times will the outer loop execute?**

```
for (int i = 1; i <= 5; i++)      {
   for (int j = i; j <= 5; j++)    {
      System.out.print("*");    }
   System.out.println();        }
```

(a) 5 (b) 15 (c)10 (d)25

(x) **Assertion(A): Method overloading in Java allows multiple methods with the same name but different parameters.**

Reason(R): Java supports method overloading through inheritance only.

(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

Question 2

i. **Write the Java expression for** [2]

$$\frac{a^x + b^y}{\sqrt[3]{x} + \sqrt[3]{y}}$$

ii. **Identify the errors and rewrite the code.** [4]

```
class A                          {
    int squareroot (int sq)       {
          return Math.sqrt(sq);  }

     static void add( int a, int b)     {
           return a+b;               }

     public static void main ( ) {
            A obj =new A( );
            double s= squareroot(9);
             add(3,5);                } }
```

iii. **Write a statement to convert a String object to double data type.** [2]

iv. Write a method that accepts a character and checks if it is a whitespace or not. [2]

Question 3 **[10]**

Riverside transport company charges for the parcels of its customers as per the following specifications given below:

Class Name : Riverside

Member variables :

String name : to store the name of the customer

double w : to store the weight of the parcel in kg

double charge : to store the charge of the parcel

Member functions :

void accept( ) : to accept the name of the customer, weight of the parcel from the user

void calculate() :

Weight in kg	Charge per kg
Upto 10kgs	₹ 25 per kg
Next 20kgs	₹ 20 per kg
Above 30kgs	₹ 10 per kg

A surcharge of 5% is charged on the bill.

void print( ) : to print the name of the customer ,weight of the parcel, total bill inclusive of surcharge in a tabular form in the following format:

Name Weight Bill amount

Define a class with the above mentioned specifications, create the main method, create an object and invoke the member methods.

Question 4 **[10]**

Design a class to overload a function fun( ) as follows:

void fun( ): Method to display the followong pattern

1
2 4
3 5 7
6 8 10 12
9 11 13 15 17

void fun(int num): A Dudeney number is a positive integer that is a perfect cube such that the sum of its digits is equal to the cube root of the number. Write a program to input a number and check and print whether it is a Dudeney number or not.

Example:

Consider the number 512.

Sum of digits = 5 + 1 + 2 = 8 Cube root of 512 = 8

As Sum of digits = Cube root of Number hence 512 is a Dudeney number.